

Rationalising Denominators of a Fraction



When the denominator of a fraction is a surd it is usual to change the denominator to a whole number. This is called rationalising the denominator.

Remember: $\sqrt{3} \times \sqrt{3} = 3$ and $\sqrt{10} \times \sqrt{5} = \sqrt{50} = \sqrt{25} \times \sqrt{2} = 5\sqrt{2}$

Example 1. $\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{1 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{3}}{3}$

Example 2. $\frac{2\sqrt{10}}{\sqrt{5}} = \frac{2\sqrt{10}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{10} \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}} = \frac{2 \times \sqrt{10} \times \sqrt{5}}{5} = \frac{2 \times \sqrt{50}}{5} = \frac{2 \times \sqrt{25} \times \sqrt{2}}{5}$
 $= \frac{2 \times 5 \times \sqrt{2}}{5} = 2\sqrt{2}$

A. Rationalise the denominator then simplify the fraction.

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|---------------------------|----------------------------|---------------------------|----------------------------|----------------------------|-----------------------------|
| 1). $\frac{1}{\sqrt{2}}$ | 2). $\frac{1}{\sqrt{5}}$ | 3). $\frac{1}{\sqrt{7}}$ | 4). $\frac{1}{\sqrt{6}}$ | 5). $\frac{1}{\sqrt{11}}$ | 6). $\frac{1}{\sqrt{10}}$ |
| 7). $\frac{3}{\sqrt{3}}$ | 8). $\frac{5}{\sqrt{5}}$ | 9). $\frac{2}{\sqrt{2}}$ | 10). $\frac{7}{\sqrt{7}}$ | 11). $\frac{6}{\sqrt{6}}$ | 12). $\frac{11}{\sqrt{11}}$ |
| 13). $\frac{2}{\sqrt{6}}$ | 14). $\frac{2}{\sqrt{10}}$ | 15). $\frac{3}{\sqrt{6}}$ | 16). $\frac{5}{\sqrt{15}}$ | 17). $\frac{7}{\sqrt{14}}$ | 18). $\frac{3}{\sqrt{21}}$ |
| 19). $\frac{4}{\sqrt{2}}$ | 20). $\frac{15}{\sqrt{5}}$ | 21). $\frac{6}{\sqrt{3}}$ | 22). $\frac{18}{\sqrt{6}}$ | 23). $\frac{35}{\sqrt{5}}$ | 24). $\frac{14}{\sqrt{7}}$ |

B. Express each of the following in simplest form with a rational denominator.

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|------------------------------------|------------------------------------|------------------------------------|------------------------------------|---|---|
| 1). $\frac{6}{\sqrt{8}}$ | 2). $\frac{8}{\sqrt{12}}$ | 3). $\frac{6}{\sqrt{24}}$ | 4). $\frac{9}{\sqrt{18}}$ | 5). $\frac{4}{\sqrt{20}}$ | 6). $\frac{8}{\sqrt{32}}$ |
| 7). $\frac{\sqrt{12}}{\sqrt{3}}$ | 8). $\frac{\sqrt{18}}{\sqrt{2}}$ | 9). $\frac{\sqrt{75}}{\sqrt{3}}$ | 10). $\frac{\sqrt{5}}{\sqrt{20}}$ | 11). $\frac{\sqrt{3}}{\sqrt{12}}$ | 12). $\frac{\sqrt{3}}{\sqrt{6}}$ |
| 13). $\frac{2\sqrt{8}}{\sqrt{32}}$ | 14). $\frac{3\sqrt{5}}{\sqrt{15}}$ | 15). $\frac{5\sqrt{7}}{\sqrt{35}}$ | 16). $\frac{2\sqrt{5}}{\sqrt{10}}$ | 17). $\frac{\sqrt{3}\sqrt{5}}{\sqrt{30}}$ | 18). $\frac{\sqrt{2}\sqrt{3}}{\sqrt{18}}$ |
| 19). $\frac{7}{2\sqrt{3}}$ | 20). $\frac{\sqrt{8}}{2\sqrt{5}}$ | 21). $\frac{\sqrt{10}}{3\sqrt{2}}$ | 22). $\sqrt{\frac{9}{10}}$ | 23). $\sqrt{\frac{4}{3}}$ | 24). $\sqrt{\frac{25}{8}}$ |

C.

- 1). Write $\frac{21}{\sqrt{7}}$ in the form $a\sqrt{7}$ where a is a number to be found.
- 2). Write $\frac{10}{\sqrt{5}}$ in the form $a\sqrt{3}$ where a is a number to be found.
- 3). Write $\frac{5}{\sqrt{3}}$ in the form $\frac{a\sqrt{3}}{b}$ where a and b are numbers to be found.
- 4). Write $\frac{3}{\sqrt{5}}$ in the form $\frac{a\sqrt{5}}{b}$ where a and b are numbers to be found.
- 5). Rationalise the denominator and simplify $\frac{3}{\sqrt{8}}$.
- 6). Rationalise the denominator and simplify $\frac{6}{\sqrt{2}}$.
- 7). Expand and simplify $\sqrt{6}(5 - \sqrt{6})$.
- 8). Expand and simplify $(2 + \sqrt{3})(2 - \sqrt{3})$.

